

YAHYA MENKARI

AI Engineering | Machine Learning | RAG Systems | Big Data

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PROFESSIONAL SUMMARY

Final-year Big Data & Artificial Intelligence Engineering student with hands-on experience in Machine Learning, Retrieval-Augmented Generation (RAG), Graph Databases, and Cloud-based AI systems. Developed AI-powered applications using Neo4j, Qdrant, Gemini, Google Cloud, and modern ML techniques. Currently conducting research on weather forecasting using machine learning and deep learning models.

EDUCATION

International University of Rabat (UIR), Morocco

- M.Eng. in Computer Science – Big Data & Artificial Intelligence (2021 – 2026)

Technical University of Sofia, Bulgaria

- Exchange Program – IT for Business Management (2025 – 2026)
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PROFESSIONAL EXPERIENCE

Technical University of Sofia, Bulgaria

Machine Learning Research Intern | Feb 2026 – Jun 2026

- Developed machine learning and deep learning models for weather forecasting using historical meteorological datasets.
- Designed preprocessing and feature engineering pipelines including lag variables and rolling statistical features.
- Implemented and compared XGBoost, ARIMA, SARIMA, LSTM, GRU, and Temporal Convolutional Network (TCN) models.
- Evaluated forecasting performance using MAE and RMSE, metrics.
- Conducted research on state-of-the-art AI weather forecasting architectures including GenCast and WeatherNext.

Archipel Cognitive, Morocco

AI Intern | Jul 2024 – Sep 2024

- Implemented and evaluated Retrieval-Augmented Generation (RAG) architectures for AI applications.
 - Applied machine learning system design principles to improve retrieval quality and response accuracy.
 - Conducted research on vector search and knowledge-enhanced AI systems.
 - Worked with modern AI development methodologies for production-oriented environments.
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TECHNICAL SKILLS

- **Programming:** Python, Java, SQL, C
- **Machine Learning:** Scikit-Learn, XGBoost, Random Forest, Logistic Regression, KNN, SVM
- **Deep Learning:** LSTM, GRU, TCN, CNN
- **LLM & RAG:** Gemini, CrewAI, Neo4j, Qdrant, FAISS, Elasticsearch
- **Cloud & DevOps:** Google Cloud Platform, Firestore, Docker, Microsoft Azure
- **Data & Analytics:** MongoDB, Power BI, ETL Pipelines, Statistical Analysis

PROJECTS

Graph-Based AI Symptom Checker

- Designed a medical knowledge graph using Neo4j to model diseases, symptoms, medications, and their relationships.
- Integrated PubMed data and Qdrant vector search for semantic medical information retrieval.
- Implemented Retrieval-Augmented Generation (RAG) techniques to improve answer quality and contextual relevance.

Medical AI Assistant with Specialist Referral System

- Developed a multi-agent healthcare assistant using CrewAI and Gemini models.
- Automated specialist referral, information retrieval, and report generation workflows.
- Built an interactive Streamlit interface supporting patient guidance and appointment recommendations.

AI-Driven Prescription Reader

- Developed an OCR-based application for handwritten prescription recognition.
- Used Google Cloud Vision API for text extraction and Firestore for medication validation.
- Implemented multilingual medication information retrieval and recommendation features.

CERTIFICATIONS

- Building Agentic RAG with LlamaIndex — DeepLearning.AI
- Microsoft Azure AI Engineer Associate (In Progress)

LANGUAGES

Arabic (Native) | French (B2) | English (B2)